

← Main menu

Build Custom Processors with Document AI

Course · 6 hours 15 minutes · 50% complete

Course overview

Build Custom Processors with Document AI

✓ Optical Character Recognition (OCR) with Document AI (Python)

✓ Form Parsing with Document AI (Python)

✓ Using Specialized Processors with Document AI (Python)

Uptraining with Document AI Workbench

Custom Document Extraction with Document AI Workbench

Course > Build Custom Processors with Document AI >

Quick tip: Review the prerequisites before you run the lab

End Lab

00:12:49

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Username

student-02-dac8d09491ai

Copy icon

Password

WeF1V0ckaQtH

Copy icon

Project ID

qwiklabs-gcp-02-c7b4ff3i

Copy icon

< Previous

Next >

Uptraining with Document AI Workbench

Lab

1 hour

No cost

Intermediate

★★★★★

ⓘ

 This lab may incorporate AI tools to support your learning.

GSP1141

Google Cloud Self-Paced Labs

Lab instructions and tasks

90/100

GSP1141

Overview

Objectives

Setup and requirements

Task 1. Enable the Document AI API

Task 2. Create a processor

Task 3. Create a dataset

Task 4. Import a test document

Task 5. Label the test document

Task 6. Assign document to training set

Task 7. Import pre-labeled data

Task 8. Edit labels

Overview

[Document AI](#) is a document understanding solution that takes unstructured data (e.g. documents, emails, invoices, forms, etc.) and makes the data easier to understand, analyze, and consume. The API provides structure through content classification, entity extraction, advanced searching, and more. With Document AI Workbench, you can achieve higher document processing accuracy by creating fully customized models using your own training data.

By using [uptraining](#) through Document AI Workbench, you can achieve higher document processing accuracy by providing additional labeled examples for Specialized Document Types and creating a new model version. In this lab, you will create an Invoice Parser processor, configure the processor for uptraining, label example documents, and uptrain the processor.

The document dataset used in this lab consists of randomly-generated invoices for a fictional piping company.

Objectives

In this lab, you will learn how to perform the following tasks:

- Configure Uptraining for an Invoice Parser processor.
- Label Document AI training data using the annotation tool.
- Kick off training for a new model version.

Setup and requirements

Before you click the Start Lab button

Read these instructions. Labs are timed and you cannot pause them. The timer, which starts when you click **Start Lab**, shows how long Google Cloud resources are made available to you.

This hands-on lab lets you do the lab activities in a real cloud environment, not in a simulation or demo environment. It does so by giving you new, temporary credentials you use to sign in and access Google Cloud for the duration of the lab.

To complete this lab, you need:

- Access to a standard internet browser (Chrome browser recommended).

Note: Use an Incognito (recommended) or private browser window to run this lab. This prevents conflicts between your personal account and the student account, which may cause extra charges incurred to your personal account.

- Time to complete the lab—remember, once you start, you cannot pause a lab.

Note: Use only the student account for this lab. If you use a different Google Cloud account, you may incur charges to that account.

How to start your lab and sign in to the Google Cloud console

to select your payment method. On the left is the Lab Details pane with the following:

- The Open Google Cloud console button
- Time remaining
- The temporary credentials that you must use for this lab
- Other information, if needed, to step through this lab

2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window** if you are running the Chrome browser).

The lab spins up resources, and then opens another tab that shows the Sign in page.

Tip: Arrange the tabs in separate windows, side-by-side.

Note: If you see the **Choose an account** dialog, click **Use Another Account**.

3. If necessary, copy the **Username** below and paste it into the **Sign in** dialog.

student-vz-qac0009491a@qwiklabs.net

You can also find the Username in the Lab Details pane.

4. Click **Next**.

5. Copy the **Password** below and paste it into the **Welcome** dialog.

WeF1V0ckaQtH

You can also find the Password in the Lab Details pane.

6. Click **Next**.

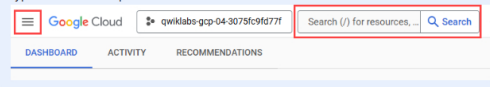
Important: You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.

Note: Using your own Google Cloud account for this lab may incur extra charges.

- Accept the terms and conditions.
- Do not add recovery options or two-factor authentication (because this is a temporary account).
- Do not sign up for free trials.

After a few moments, the Google Cloud console opens in this tab.

Note: To access Google Cloud products and services, click the **Navigation menu** or type the service or product name in the **Search** field.



Activate Cloud Shell

Cloud Shell is a virtual machine that is loaded with development tools. It offers a persistent 5GB home directory and runs on the Google Cloud. Cloud Shell provides

1. Click **Activate Cloud Shell**  at the top of the Google Cloud console.

2. Click through the following windows:

- Continue through the Cloud Shell information window.
- Authorize Cloud Shell to use your credentials to make Google Cloud API calls.

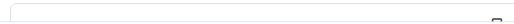
When you are connected, you are already authenticated, and the project is set to your **Project_ID**, `qwiklabs-gcp-02-c7b4ff3a41ea`. The output contains a line that declares the **Project_ID** for this session:

```
Your Cloud Platform project in this session is set to qwiklabs-gcp-02-c7b4
```



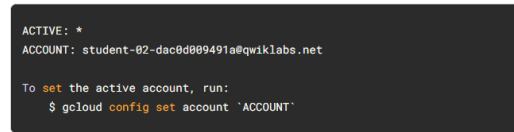
gcloud is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

3. (Optional) You can list the active account name with this command:



4. Click **Authorize**.

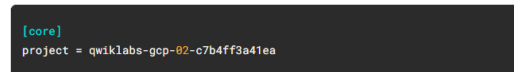
Output:



5. (Optional) You can list the project ID with this command:



Output:



[overview guide.](#)

Task 1. Enable the Document AI API

Before you can begin using Document AI, you must enable the API.

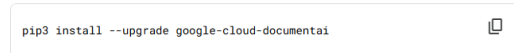
1. Open Cloud Shell by clicking the **Activate Cloud Shell** button at the top of the console.
2. In Cloud Shell, run the following command to enable the API for Document AI.



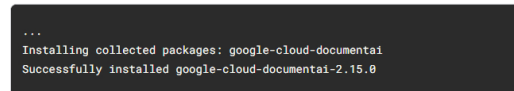
You should see something like this:



3. Run the following command to install the Python client libraries for Document AI.

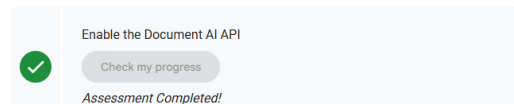


You should see something like this:



Now, you're ready to use the Document AI API!

Click **Check my progress** to verify the objective.

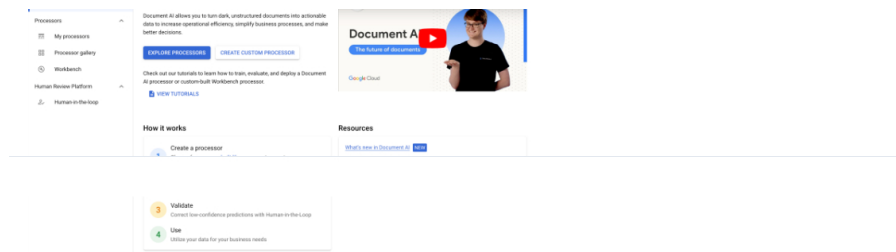


Task 2. Create a processor

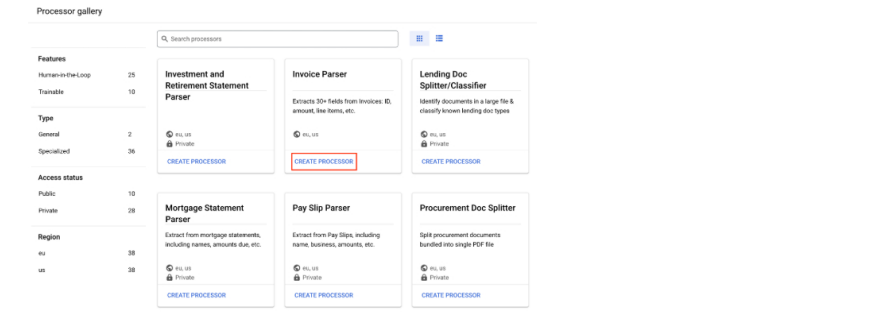
You must first create a Form Parser processor instance to use in the Document AI Platform for this tutorial.

1. From the Navigation Menu, under **Artificial Intelligence**, select **Document AI**.

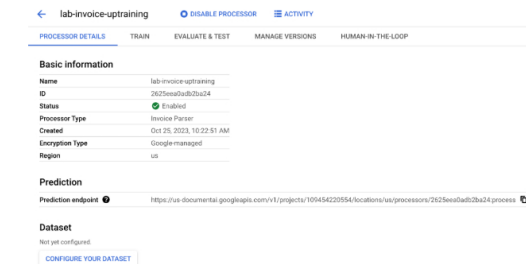




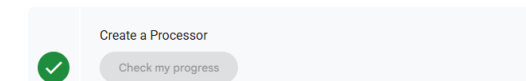
2. Click **Explore Processors**, scroll down to **Specialized** and inside **Invoice Parser**, click **Create Processor**.



4. Click **Create** to create your processor.



Click **Check my progress** to verify the objective.



Task 3. Create a dataset

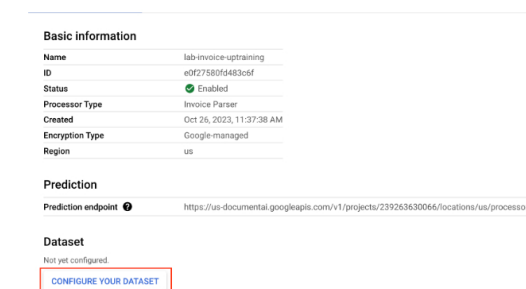
In order to train your processor, we will have to create a dataset with training and testing data to help the processor identify the entities we want to extract.

You will need to create a new bucket in Cloud Storage to store the dataset.

1. Open Cloud Shell and run the following commands to create a bucket.



2. Go to the **Processor Details** tab, and under **Dataset**, click on **Configure Your Dataset**.



3. Click on **Show Advanced Options**, and select **I'll specify my own storage location**.

PROCESSOR DETAILS

TRAIN

EVALUATE & TEST

MANAGE VERSIONS

HUMAN-IN-THE-LOOP

Configure dataset

Storage location

Choose an empty folder to store them.

gs:// Dataset folder *

BROWSE

Select an empty GCS folder with no existing data or documents

SHOW LESS

CONTINUE

4. Paste the bucket name from the bucket you created in step one into the **Destination Path** field. (Don't include `gs://`).

This should be: `qwiklabs-gcp-02-c7b4ff3a41ea-uptraining-lab`.

Select an empty folder on Cloud Storage to store your dataset

You can import your documents after the dataset is created.

Destination path *

gs://

qwiklabs-gcp-01-798ec90df88f-uptraining-lab/

BROWSE

CREATE DATASET

CANCEL

5. Click **Continue** or click **Create Dataset**.

6. Wait for the dataset to be created, then it should direct you to the Processor Details page. Your dataset details should be listed under **Dataset**.

lab-invoice-uptraining

DISABLE PROCESSOR

ACTIVITY

PROCESSOR DETAILS

TRAIN

EVALUATE & TEST

MANAGE VERSIONS

HUMAN-IN-THE-LOOP

Name

lab-invoice-uptraining

ID

ed9048ca18c7fed1

Status

Enabled

Processor Type

Invoice Parser

Encryption Type

Google managed key

Prediction

Prediction endpoint

https://us-documentai.googleapis.com/v1/projects/544769099008/locations/us/processors/ed9048ca18c7fed1/process

Dataset

Storage

gs://qwiklabs-gcp-01-798ec90df88f-uptraining-lab/

Total document

0

Training document

0

Test document

0

Unsigned document

0

Click **Check my progress** to verify the objective.

Create a Dataset

Assessment Completed!

Task 4. Import a test document

Now you will import a sample invoice PDF into your dataset.

- Go to the **Train** tab, then click on **Import Documents**.
- A sample PDF is available for you to use in this lab. Copy and paste the following link into the **Source Path** box. Leave the **Data split** as *Unsigned* for now.

c:\cloud-samples-data\documentai\code\labs\uptraining\pdfs

the-Loop data). [Learn more](#) about supported formats and other import requirements.

Select a source folder on Cloud Storage

Enter the Cloud Storage path to the folder(s) where your documents are stored. Your documents will be recursively imported.

Source path *

gs://

cloud-samples-data/documentai/code/

BROWSE

Data split

Unsigned

ADD ANOTHER FOLDER

Auto-labeling

Get a head start on labeling your documents using predictions from your processor's existing versions. [See example](#)

☐ Import with auto-labeling

3. Click **Import**.



Assessment Completed!

Next, you will identify text elements and labels for the entities you would like to extract. These labels will be used to train your model to parse this specific document structure and identify the correct types.

- 
- A screenshot of the Microsoft Word application window. The top ribbon is visible, and the 'Insert' tab is selected, highlighted with a red rectangle. The ribbon includes icons for Insert, Font, Layout, References, Mailings, Review, and Send. The main document area is blank.

- Highlight the text "14368 Pipeline Ave Chino, CA 91710" and assign the label `supplier_address`.

McWilliam Piping
International Piping Company
4388 Pipeline Ave Chino, CA 91719

Invoice: 10001
Due Date: 2020-01-02

Item			Cost
Knuuckle Couplers			\$669.87
PVC Pipe 12 Inch	7	\$15.90	\$111.30
Copper Pipe	7	\$91.20	\$638.40
		Subtotal	\$1,411.57
		Tax	\$113.57

		Total	\$1,533.14
--	--	--------------	------------

3. Highlight the text "10001" and assign the label `invoice_id`.

McWilliam Piping
International Piping Company
14368 Pipeline Ave Chino, CA 91710

Invoice: 10001
Due Date: 2020-01-02

Item	Quantity	Price	Cost
Knuckle Couplers	9	\$74.43	\$669.87
PVC Pipe 12 Inch	7	\$15.90	\$111.30
Copper Pipe	7	\$91.20	\$638.40
		Subtotal	\$1,419.57
		Tax	\$113.57
		Total	\$1,533.14

McWilliam Piping
International Piping Company
14368 Pipeline Ave Chino, CA 91710

Invoice: 10001
Due Date: 2020-01-02

Item	Quantity	Price	Cost
Knuckle Couplers	9	\$74.43	\$669.87
PVC Pipe 12 Inch	7	\$15.90	\$111.30
Copper Pipe	7	\$91.20	\$638.40
		Subtotal	\$1,419.57
		Tax	\$113.57
		Total	\$1,533.14

4. Highlight the text "Knuckle Couplers" and assign the label `line_item/description`.

McWilliam Piping
International Piping Company
14368 Pipeline Ave Chino, CA 91710

Invoice: 10001
Due Date: 2020-01-02

Item	Quantity	Price	Cost
Knuckle Couplers	9	\$74.43	\$669.87
PVC Pipe 12 Inch	7	\$15.90	\$111.30
Copper Pipe	7	\$91.20	\$638.40
		Subtotal	\$1,419.57
		Tax	\$113.57
		Total	\$1,533.14

6. Highlight the text "9" and assign the label `line_item/quantity`. Assign it to `line_item` (Knuckle Couplers).

McWilliam Piping
International Piping Company
14368 Pipeline Ave Chino, CA 91710

Invoice: 10001
Due Date: 2020-01-02

Item	Quantity	Price	Cost
Knuckle Couplers	9	\$74.43	\$669.87

			Total	\$1,533.14
--	--	--	--------------	------------

McWilliam Piping

International Piping Company

14368 Pipeline Ave Chino, CA 91710

Invoice: 10001
Due Date: 2020-01-02

Item	Quantity	Price	Cost
Knuckle Couplers	9	\$2649.87	
PVC Pipe 12 Inch	7	\$191.20	
Copper Pipe	7	\$638.40	

		Subtotal	01,419.57
		Tax	0113.57
		Total	01,533.14

McWilliam Piping
International Piping Company
14388 Pipeline Ave. Clero, CA 91716

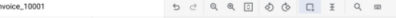
Invoice: 10001
Due Date: 2020-01-02

Item	Quantity	Price	Cost
Knuckle Couplers	9	\$74.43	\$669.87
PPC Pipe 12 Inch	7	\$15.90	\$111.30

		Subtotal	01,419,97
		Tax	0113,57
		Total	01,533,54

line_item		(Knuckle Couplers 74.43 66)			
line_item/amount	669.87	Add more rows			
line_item/description	Knuckle Couplers				
line_item/product_code	--				
line_item/purchase or	--				
line_item/quantity	9				
line_item/unit	--				

It should look like this when complete:



The screenshot shows a Google Cloud Storage bucket named 'fake_invoice_10001'. The bucket contains three objects: 'line_item.json' (5569 B), 'line_item_description' (Kuduise Couplings), and 'line_item_product_code'. The 'line_item.json' object is highlighted. To the right, a preview of the 'line_item.json' file is shown, displaying the company name 'McWilliam Piping International Piping Company' in blue text.

12. Highlight the text "1,419.57" (next to Subtotal) and assign the label `net_amount`.

13. Highlight the text "113.57" (next to Tax) and assign the label `total_tax_amount`.

14. Highlight the text "1,533.14" (next to Total) and assign the label `total_amount`.

15. Highlight one of the "\$" characters and assign the label `currency`.

		Total	\$1,933.14
--	--	-------	------------

fake_invoice_10001

Filter Type to filter

Show empty fields

amount_due

amount_paid,invoice_total

carrier

currency \$

currency_exchange_rate

customer_ref_id

customer_ref_id

delivery_date

due_date 2020-01-01

freight_amount

invoice_date

invoice_ref

Invoice: 0000

Due Date: 2020-01-01

McWilliam Piping
International Pty Company
www.mcwilliam.com.au

Item	Quantity	Price	Cost
McWilliam Piping International Pty Company	10	100.00	1000.00
PIP Pipe 12 inch	10	115.00	1150.00
McWilliam Piping International Pty Company	10	100.00	1000.00

17. Here is the full list of labels and values

line_item/unit_price	15.90
line_item/amount	111.30
line_item/description	Copper Pipe
line_item/quantity	7
line_item/unit_price	91.20
line_item/amount	638.40
net_amount	1,419.57
total_tax_amount	113.57
total_amount	1,533.14
currency	\$

 Create Labels [Check my progress](#)

[← lab-invoice-uptraining](#)
[DISABLE PROCESSOR](#)
[ACTIVITY](#)

[PROCESSOR DETAILS](#)
[TRAIN](#)
[EVALUATE & TEST](#)
[MANAGE VERSIONS](#)
[HUMAN-IN-THE-LOOP](#)

All documents

1

1 selected

ASSIGN TO SET

DELETE

Labeled

1

Select

Training

Auto-labeled

0

all

Test

Unlabeled

0

Unassigned

Training

0

Test

0

Unassigned

1

Filter

Filter Labels

+

fake_invoice_10001

1 page

- Notice the Data Split numbers have changed. You should see 1 document in the Training set and 0 in the Test set.

Click **Check my progress** to verify the objective.

✓

Assign Document to Training Set

Check my progress

Assessment Completed!

Document AI Uptaining requires a minimum of 10 documents in both the training and test sets, along with 10 instances of each label in each set. It's recommended to have at least 50 documents in each set with 50 instances of each label for best performance. More training data generally equates to higher accuracy.

It will take a long time to manually label 100 documents, so we have some pre-labeled documents that you can import for this lab. You can import pre-labeled document files in the [Document .json](#) format. These can be results from calling a processor and verifying the accuracy using [Human in the Loop \(HITL\)](#).

Note: when importing pre-labeled data, it is **highly recommended** to manually review annotations before a model is trained.

- In the training management console, click on **Import Documents**.
- Enter the following path in **Source path**. This bucket contains pre-labeled documents in the Document JSON format.

```
c:\cloud-samples-data\documentai\Custom\Invoices\JSON
```

documents to have 80% in the training set, and 20% in the test set. Leave **Import with auto-labeling** unchecked.

- Click **Import**. The import might take several minutes to complete.

When the import is finished, you will find the documents on the Train tab.

lab-invoice-uptaining

DISABLE PROCESSOR

ACTIVITY

LEARN

PROCESSOR DETAILS

TRAIN

EVALUATE & TEST

MANAGE VERSIONS

HUMAN-IN-THE-LOOP

All documents

201

Filter

Enter label or property name

Labeled

201

Select all

Auto-labeled

0

Unlabeled

0

IMPORT DOCUMENTS

Data split

Training

151

Test

50

Unassigned

0

Filter

Filter Labels

+

EDIT SCHEMA

Label

↑

Instances

amount_due

0

amount_paid_since_last_invoice

0

carrier

0

currency

1

Training

Use this dataset to train a new processor version for improved performance

UPGRADE AS A VERSION

Labeling tasks

If your dataset still needs to be labeled, create a labeling task to have human specialists label it for you

CREATE LABELING TASK

Auto-label documents

Use your processor's model to save time on labeling unlabeled documents

AUTO-LABEL DOCUMENTS

Label stats

Check the status of your dataset to see if you're ready for training

Click **Check my progress** to verify the objective.

✓

Import pre-labeled data

Check my progress

Assessment Completed!

Task 8. Edit labels

2. You should now be in the Label Management console.

3. Use the Checkboxes and the **Disable/Enable** buttons to mark **ONLY** the following labels as **Enabled**.

- receiver_address
- receiver_name
- supplier_address
- supplier_name
- total_amount

4. Click **Save** when finished.

- Click on the Back arrow to return to the Dataset management console. If asked to save the schema, click **Save**.

Notice that the labels with 0 instances have been marked as Inactive.

Click **Check my progress** to verify the objective.

When importing unlabeled documents for a processor with an existing deployed processor version, you can use [Auto-labeling](#) to save time on labeling.

1. On the **Train** page, Click **Import Documents**.
2. Copy and paste the following storage path. This directory contains 5 unlabeled invoice PDFs. From the **Data split** dropdown list, select **Training**.

☐

3. In the **Auto-labeling** section, select the **Import with auto-labeling** checkbox.

- For example: `pretrained-invoice-v1.4-2022-10-21`

5. Click **Import** and wait for the documents to import. You can leave this page and return later.

- When complete, the documents appear in the **Train** page in the **Auto-labeled** section. You cannot use auto-labeled documents for training or testing without marking them as labeled.

6. Click the **Auto-labeled** section to view the auto-labeled documents.

The screenshot shows the 'lab-invoice-uptraining' page in the AWS SageMaker console. The page is divided into several sections: 'PROCESSOR DETAILS', 'TRAIN', 'EVALUATE & TEST', 'MANAGE VERSIONS', and 'HUMAN-IN-THE-LOOP'. The 'TRAIN' section displays a table of documents with the following data:

Document Type	Count
All documents	231
Labeled	231
Auto-labeled	5
Unlabeled	0

The 'EVALUATE & TEST' section includes a filter dropdown menu with the following options: 'Filter', 'Labeled (231)', 'Auto-labeled', and 'Enter label or property name'. The 'Labeled (231)' option is currently selected. Below the filter, there is a 'Select all' checkbox and a grid of document thumbnails.

Data split	
Training	156
Test	93
Unassigned	0

Label	Instances
amount_date	0
amount_paid_amount_text_phrase	0
currency	1

7. Click the first document to enter the labeling console.

8. Verify the labels, bounding boxes, and values to ensure they are correct. Label any values that were omitted. Click **Mark as labeled** when finished with each document.

9. Repeat the label verification for each auto-labeled document, then return to the **Train** page to use the data for training.

Click **Check my progress** to verify the objective.

Auto-label newly imported documents

[Check my progress](#)

Assessment Completed!

Task 10. Uptrain the Model

Now, you are ready to begin training your Invoice Parser.

1. Click **Uptrain New Version**.

Tab-instance-uptraining

ENABLE PROCESSOR

ACTIVITY

PROCESSOR DETAILS

TASK

EVALUATE & TEST

MANAGE VERSIONS

MANAGE IN THE LOOP

all documents

206

Filter

Filter value or property name

Labelled

183

Select all

Unlabelled

0

EXPORT DOCUMENTS

Data split

156

Training

93

Unassigned

0

EDIT SCREEN

Label

Instances

amount_date

0

amount_paid_amount_text_phrase

0

currency

1

Training

Use this dataset to train a new processor version for document classification

UPTRAIN NEW VERSION

Labeling tasks

If your dataset still needs to be labeled, create a labeling task & have better supervision data for the job

CREATE LABELING TASK

Auto-label documents

Use your processor's labels to auto-label unlabelled documents

AUTO LABEL DOCUMENTS

Label status

Check the status of your processor in case it needs more data

model version this new version will be built from. If you're using a new processor, the only option should be **Google Pretrained Next with Uptraining**.

Train new version

You're ready to train. Your dataset meets the recommended number of instances assigned to each label.

Version name *

code-lab-uptraining-test-1

Must start with a letter. Can use letters, numbers, spaces, dashes, and underscores.

Base version *

Google Pretrained Next with Uptraining

[START TRAINING](#)

[VIEW LABEL STATS](#)

[CANCEL](#)

3. Click on **Start Training** to begin the Uptraining process. You should be redirected to the Dataset management page. You can view the training status on the right side. Training will take a few hours to complete.

your dataset.

Labeled documents

Labels	Documents	Training	Test
invoice_date	183	146	37
line_item	206	166	40
amount	206	166	40
description	206	166	40
receiver_address	200	161	39
receiver_name	202	162	40
supplier_address	166	135	31
supplier_name	205	165	40

Uptrain the Model

Check my progress

5. If you click on the version name, you will be directed to the **Manage Versions** page, which shows the Version ID and the current status of the Training Job.

Great! You have now started training your first uptraining job for Document AI. Since the training job will take around an hour, the lab will end here. If you are interested in learning more about testing the model version and evaluation, you can check out the following [documentation guide](#).

Congratulations!

Congratulations, you've successfully used Document AI to uptrain an Invoice Parser. You

the processing response.

Next steps/Learn more

Check out the following resources to learn more about Document AI and the Python Client Library:

- [The Future of Documents - YouTube Playlist](#)
- [Document AI Documentation](#)
- [Document AI Python Client Library](#)
- [Document AI Samples](#)

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